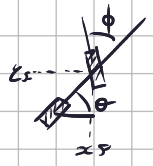
1)



$\begin{pmatrix} -\sin(\theta+\phi) & \cos(\theta+\phi) \end{pmatrix}\begin{pmatrix} \dot{x}_s \\ \dot{\xi}_s \end{pmatrix} = 0 \Rightarrow -\dot{x}_s s_{\theta\phi} + \dot{\xi}_s c_{\theta\phi} = 0$ The other is

$\begin{pmatrix} -s_\theta & c_\theta \end{pmatrix}\begin{pmatrix} \dot{x} \\ \dot{\xi} \end{pmatrix} = 0$ where $\begin{cases} x = x_s - \ell c_\theta \\ \xi = \xi_s - \ell s_\theta \end{cases} \Rightarrow \begin{cases} \dot{x} = \dot{x}_s + \dot{\theta}\ell s_\theta \\ \dot{\xi} = \dot{\xi}_s - \dot{\theta}\ell c_\theta \end{cases}$

$\Rightarrow -s_\theta(\dot{x}_s + \dot{\theta}\ell s_\theta) + c_\theta(\dot{\xi}_s - \dot{\theta}\ell c_\theta) = 0$

$\Rightarrow -\dot{\theta}\ell - \dot{x}_s s_\theta + \dot{\xi}_s c_\theta = 0$ hence

$$\underbrace{\begin{pmatrix} -s_{\theta\phi} & c_{\theta\phi} & 0 & 0 \\ -s_\theta & c_\theta & -\ell & 0 \end{pmatrix}}_{A}\begin{pmatrix} \dot{x}_s \\ \dot{\xi}_s \\ \dot{\theta} \\ \dot{\phi} \end{pmatrix} = 0$$

$\{g_1, g_2\}$ = $\mathcal{B}$ We look for $\mathcal{B}$ s.t. $\operatorname{span}\mathcal{B} = \mathcal{N}(A)$

$\Rightarrow g_2 = (0 \;\; 0 \;\; 0 \;\; 1)^T$ (ϕ is free) $\quad g_1 = (c_{\theta\phi} \;\; s_{\theta\phi} \;\; \alpha \;\; 0)^T \Rightarrow$

$$Ag_1 = \begin{cases} 0 \\ -s_\theta c_{\theta\phi} + c_\theta s_{\theta\phi} - \ell\alpha = s_\phi - \ell\alpha = 0 \Rightarrow \alpha = s_\phi/\ell \end{cases}$$

The model is:

$$\begin{cases} \dot{x}_s = c_{\theta\phi}\, v \\ \dot{\xi}_s = s_{\theta\phi}\, v \\ \dot{\theta} = s_\phi/\ell\, v \\ \dot{\phi} = \omega \end{cases}$$

To prove controllability, we consider the accessibility distribution.

$$g_3 = [g_1, g_2] = \underbrace{\frac{\partial g_2}{\partial q} g_1}_{0} - \frac{\partial g_1}{\partial q} g_2 = -\begin{pmatrix} 0 & 0 & -s_{\theta\phi} & -s_{\theta\phi} \\ 0 & 0 & c_{\theta\phi} & c_{\theta\phi} \\ 0 & 0 & 0 & c_\phi/\ell \\ 0 & 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -s_{\theta\phi} \\ c_{\theta\phi} \\ c_\phi/\ell \\ 0 \end{pmatrix}$$

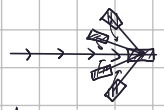
g_3 is lin. independent from g_1, g_2, hence $\operatorname{rank}\Delta_k \geq 3$ (Δ_k := access. dist.)

Consider $g_4 = [g_2, g_3] = \frac{\partial g_3}{\partial q} g_2 - \underbrace{\frac{\partial g_2}{\partial q} g_3}_{0} = \begin{pmatrix} 0 & 0 & -c_{\theta\phi} & -c_{\theta\phi} \\ 0 & 0 & -s_{\theta\phi} & -s_{\theta\phi} \\ 0 & 0 & 0 & -s_\phi/\ell \\ 0 & 0 & 0 & 0 \end{pmatrix} g_2 = \begin{pmatrix} -c_{\theta\phi} \\ -s_{\theta\phi} \\ -s_\phi/\ell \\ 0 \end{pmatrix} = -g_1$

$$g_5 = [g_1, g_3] = \begin{pmatrix} 0 & 0 & -c_{\theta\phi} & -c_{\theta\phi} \\ 0 & 0 & -s_{\theta\phi} & -s_{\theta\phi} \\ 0 & 0 & 0 & -s_\phi/\ell \\ 0 & 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} c_{\theta\phi} \\ s_{\theta\phi} \\ s_\phi/\ell \\ 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & -s_{\theta\phi} & -s_{\theta\phi} \\ 0 & 0 & c_{\theta\phi} & c_{\theta\phi} \\ 0 & 0 & 0 & c_\phi/\ell \\ 0 & 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} -s_{\theta\phi} \\ c_{\theta\phi} \\ c_\phi/\ell \\ 0 \end{pmatrix} \qquad * \; n := \frac{1}{\ell}$$

$$g_5 = \begin{cases} -C_{\theta\phi} S_\phi n - S_{\theta\phi} C_\phi n \\ -S_{\theta\phi} S_\phi n + C_{\theta\phi} C_\phi n \\ 0 \\ 0 \end{cases} = \begin{cases} -n \sin(\theta + 2\phi) \\ n \cos(\theta + 2\phi) \\ 0 \\ 0 \end{cases}$$

We see that $\text{rank}\begin{pmatrix} g_1 & g_2 & g_3 & g_5 \end{pmatrix} = 4 \Rightarrow$ the system is controllable.

Are (x_s, ξ_s) flat outputs? No, since from a trajectory of the front wheel there are an infinite amount of possible trajectories for the rear wheel: 

Consider again the system

$$\begin{cases} \dot{x}_s = C_{\theta\phi} v \\ \dot{\xi}_s = S_{\theta\phi} v \\ \dot{\theta} = n S_\phi v \\ \dot{\phi} = \omega \end{cases}$$ how to control it? Let's derive:

$$\ddot{x}_s = -(\dot{\theta} + \dot{\phi}) S_{\theta\phi} v + C_{\theta\phi} \dot{v}$$
$$\ddot{\xi}_s = (\dot{\theta} + \dot{\phi}) C_{\theta\phi} v + S_{\theta\phi} \dot{v}$$

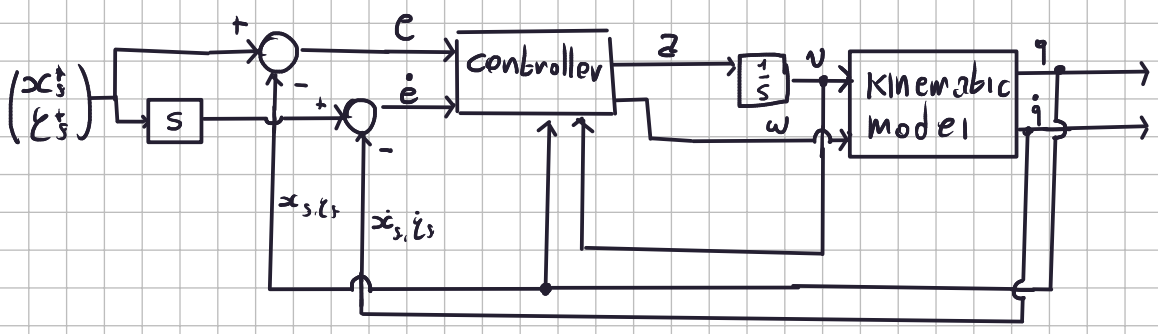
$\tilde{q} = (q, v)$ — v becomes a state

$a := \dot{v}$ is our input

$$\Rightarrow \begin{cases} \ddot{x}_s = -\left(n S_\phi v + \omega\right) S_{\theta\phi} v + C_{\theta\phi} a \\ \ddot{\xi}_s = \left(n S_\phi v + \omega\right) C_{\theta\phi} v + S_{\theta\phi} a \end{cases} \Rightarrow \begin{pmatrix} \ddot{x}_s \\ \ddot{\xi}_s \end{pmatrix} = \underbrace{\begin{pmatrix} C_{\theta\phi} & -S_{\theta\phi} v \\ S_{\theta\phi} & C_{\theta\phi} v \end{pmatrix}}_{T(\tilde{q})} \underbrace{\begin{pmatrix} a \\ \omega \end{pmatrix}}_{u} + \underbrace{\begin{pmatrix} -n S_\phi S_{\theta\phi} v^2 \\ n S_\phi C_{\theta\phi} v^2 \end{pmatrix}}_{\beta(\tilde{q})}$$

We choose: $\begin{pmatrix} a \\ \omega \end{pmatrix} = T^{-1}(\tilde{q})\left(\gamma - \beta(\tilde{q})\right) \Rightarrow \begin{pmatrix} \ddot{x}_s \\ \ddot{\xi}_s \end{pmatrix} = \gamma$

We define γ as: $\gamma = \begin{cases} \gamma_1 = \ddot{x}_s^* + K_p (x_s^* - x_s) + K_D (\dot{x}_s^* - \dot{x}_s) \\ \gamma_2 = \ddot{\xi}_s^* + K_p (\xi_s^* - \xi_s) + K_D (\dot{\xi}_s^* - \dot{\xi}_s) \end{cases}$



2) $\begin{cases} \dot z_1 = v_1 \\ \dot z_2 = v_2 \\ \dot z_3 = z_2 v_1 \\ \dot z_4 = z_3 v_1 \end{cases}$ (z_1, z_4) are flat outputs, the rec. formula for v_1 is trivial: $v_1 = \dot z_1$. For z_3 is $z_3 = \frac{1}{v_1}\dot z_4 = \frac{\dot z_4}{\dot z_1}$, and $\dot z_3 = \frac{\ddot z_4}{\dot z_1} - \dot z_4 \ddot z_1 \frac{1}{\dot z_1^2} = \frac{\dot z_1 \ddot z_4 - \ddot z_1 \dot z_4}{\dot z_1^2}$

Furthermore $\dot z_3 = z_2 v_1 \Rightarrow z_2 = \frac{1}{v_1}\dot z_3 = \frac{1}{\dot z_1}\left(\frac{\dot z_1 \ddot z_4 - \ddot z_1 \dot z_4}{\dot z_1^2}\right)$

in the end $v_2 = \frac{d}{dt}\left[\frac{1}{\dot z_1}\left(\frac{\dot z_1 \ddot z_4 - \ddot z_1 \dot z_4}{\dot z_1^2}\right)\right]$

v_2 will depend also on $\ddot z_1, \ddot z_4$. To plan a trajectory from z_i, z_s, we just need to use the reconstruction formulas.

let $t \in [0,1]$

$z_1(t) = z_{1i} + t(z_{1s} - z_{1i}) \Rightarrow v_1(t) = z_{1s} - z_{1i}$

For z_4, we use a cubic:

$z_4(t) = at^3 + bt^2 + ct + d$. denote $z_{4s} - z_{4i} = \Delta_4$

$$\begin{cases} z_4(0) = d = z_{4i} \\ z_4(1) = a + b + z_{4i} = z_{4s} \\ \dot z_4(0) = c = 0 \\ \dot z_4(1) = 3a + 2b + z_{4i} = 0 \end{cases} \quad \begin{cases} d = z_{4i} \\ c = 0 \\ a = \Delta_4 - b \\ 3\Delta_4 - b + z_{4i} = 0 \end{cases} \quad \begin{cases} a = -2\Delta_4 - z_{4i} \\ b = 3\Delta_4 + z_{4i} \end{cases}$$

$$\Rightarrow z_4(t) = \underbrace{-(2\Delta_4 + z_{4i})}_{\alpha} t^3 + \underbrace{(3\Delta_4 + z_{4i})}_{\beta} t^2 + z_{4i}$$

$$\dot z_4(t) = -3\alpha t^2 + 2\beta t \qquad \ddot z_4 = -6\alpha t + 2\beta$$

$$\Rightarrow v_2 = \frac{d}{dt}\left[\frac{-6\alpha t + 2\beta}{(z_{1s} - z_{1i})^2}\right] = -\frac{6\alpha}{(z_{1s} - z_{1i})^2}$$

3) Thanks to the IMU, $\dot{x}$, $\dot{z}$ and $\dot{\theta}$ are known.

In digital control, we can integrate $\dot{q}, \ddot{q}$ to track q, with Euler integration, we can compute a first estimate of the next state as follows:

$$\begin{cases} x_{k+1} = \hat{x}_k + T_s \dot{x}_k + \frac{1}{2} T_s^2 (S_{1k} \cos\hat{\theta}_k - S_{2k} \sin\hat{\theta}_k) \\ z_{k+1} = \hat{z}_k + T_s \dot{z}_k + \frac{1}{2} T_s^2 (S_{1k} \sin\hat{\theta}_k + S_{2k} \cos\hat{\theta}_k - g) \\ \theta_{k+1} = \hat{\theta}_k + T_s \dot{\theta}_k + \frac{1}{2} T_s^2 (-d F_{2k}) \end{cases} = \hat{q}_{k+1|k} = f(\hat{q}_k)$$

($T_s \dot{\theta}_k$: exact)

We assume that on the dynamics there is a noise $v_k \sim N(0, V_k)$

Let P_k be the covariance of our estimation $\hat{q}_k$, a first estimate of P_{k+1} is:

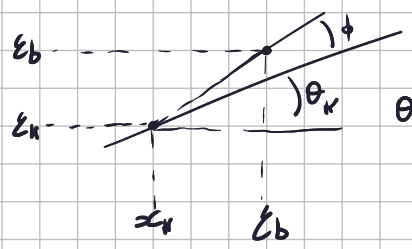
$$P_{k+1|1} = F_k P_k F_k^T + V_k \quad \text{where} \quad F_k = \frac{\partial f}{\partial q}\Big|_{q=\hat{q}_k, u=u_k}$$

$$F_k = \begin{pmatrix} 1 & 0 & (-S_{1k}\sin\hat{\theta}_k + S_{2k}\cos\hat{\theta}_k)\frac{1}{2}T_s^2 \\ 0 & 1 & (S_{1k}\cos\hat{\theta}_k - S_{2k}\sin\hat{\theta}_k)\frac{1}{2}T_s^2 \\ 0 & 0 & 1 \end{pmatrix}$$

Now we consider the measurement, given by:

$$\zeta_k = h(q_k) = \operatorname{atan2}(\zeta_b - \zeta_k, x_b - x_k) - \theta_k$$

$\zeta = z$!!



$$\theta_k + \phi = \operatorname{atan2}(\zeta_b - \zeta_k, x_b - x_k)$$

We define the innovation as $v_{k+1} = \zeta_{k+1} - h(\tilde{q}_{k+1|k})$ and we use it with the Kalman gain R_{k+1} to correct: $\tilde{q}_{k+1} = \tilde{q}_{k+1|k} + R_{k+1} v_{k+1}$.

Also: $P_{k+1} = P_{k+1|k} - R_{k+1} H_{k+1} P_{k+1|k}$ where $H_{k+1} = \frac{\partial h}{\partial q}\Big|_{q=\tilde{q}_{k+1|k}}$

$$H_{k+1} = \begin{pmatrix} -\frac{x_b - x_k}{(x_b - x_k)^2 + (\zeta_b - \zeta_k)^2} & \frac{\zeta_b - \zeta_k}{(x_b - x_k)^2 + (\zeta_b - \zeta_k)^2} & -1 \end{pmatrix}$$